The Ring of Fire

Overview

In this activity, students investigate volcanoes and the theory of plate tectonics. They will use Encarta Encyclopedia 2000 and Encarta® Interactive World Atlas to locate cities and nations along the Ring of Fire as they design guided tours of some of the volcanic regions of earth. For a related lesson on plate tectonics and earthquakes, see the lesson This Dynamic Planet.

Objectives

In completing this activity, students will:

- Locate the Ring of Fire on a world map
- Explain the origins of the region's name
- Identify nations and cities that lay along the Ring of Fire
- Link plate tectonics to the location of earth's volcanoes

Materials

Materials for making a tour booklet
Sample tour booklets (You might want to visit a travel agency for some booklets or brochures that students can use as models)
Encarta Encyclopedia 2000
Encarta Interactive World Atlas

Background Information/Fast Facts

In introducing the activity, you might want to share some of the following facts with students. (All information comes from Encarta Encyclopedia.)

- According to the Smithsonian Institution, 1,511 above-the-sea volcanoes have been active during the past 10,000 years, 539 of them erupting one or more times during written history.
- On average, 50 to 60 above-the-sea volcanoes are active in any given year. About half of these are continuations of eruptions from the previous year, and the rest are new.
- About 60 percent of the active above-the-sea volcanoes lie along the Pacific Rim.
- The United States ranks third, behind Japan and Indonesia, in the number of historically active volcanoes. When measured from its submerged base, Mauna Kea, the highest summit in Hawaii, stands 32,000 feet, or 9,700 meters tall—higher than Mt. Everest.
- As magma enters a volcanic reservoir, it may be anywhere from 1000°C (1832°F) to 2000°C (3632°F).
- Not all eruptions consist of magma or molten rock. Some eruptions consist of gases, dust, rocks, or a mixture of all three.
- When a volcano erupts, it may spew blocks of rock ("bombs") as big as a car.
- Since the beginning of recorded history, an estimated 50 volcanoes have erupted in the area that makes up the present-day United States.

Activities and Procedures

1. You might begin the lesson by sharing the Fast Facts with students. After presenting the last fact, ask students to speculate on the area of the United States with the greatest number of active volcanoes. (The nation's northwestern coast, especially the Cascades and Alaska.)
2. Have a student locate this region on a wall map of the world or in Encarta (either in the World Maps that accompany Encarta Encyclopedia or in Encarta

Interactive World Atlas). Point out the region's location along the Pacific Rim. Call on students to suggest reasons for the high level of volcanic activity in this area.

3. If students are unfamiliar with the theory of plate tectonics, you might have them read about this topic in the Encarta Encyclopedia or assign "The Changing Earth," a 1973 *National Geographic* article found in the Article Archives. (This article presents the groundbreaking work of the 1960s that revolutionized the geophysical sciences.)
4. Refer students to the map in the encyclopedia article on volcanoes. Using this map, have them explain why the Pacific Rim is sometimes called the "Ring of Fire." If you are using Encarta Interactive World Atlas, assign students to use the Map Customizer to add cities and country borders to a tectonic map showing the plates and volcanoes of earth.
5. Divide the class into groups. Direct the students to "tour" a nation or city known for its volcano or volcanoes, using their customized maps. Make sure that each group picks a different area to tour.
6. Explain that the purpose of the tour is to design a trip guide for travelers who wish to visit volcanic sites. Students will find many useful features in both the Encarta Encyclopedia and the Encarta Interactive World Atlas. The trip guides can include pictures, site locations, and more. Make sure students take advantage of the Map Treks listed in the Follow Up section of this lesson.
7. When students are done, have them exchange trip guides. Students might evaluate each other's guides, noting positive features and suggesting improvements.

Additional Activities

Tracing Volcanic Growth

Assign students to design storyboards showing the various steps in the creation of an active volcano. Explain that storyboards are like the frames in a comic strip or celluloid film. In fact, video and filmmakers often use storyboards to present a concept or story line as part of their development process. As a reference, tell students to look at the animated clip **Formation and Eruption of a Volcano** in Encarta Encyclopedia. When students are finished, have them present their storyboards to the class. Ask other students to watch the animated clip to see how closely students captured the sequence of events.

Distinguishing Types of Volcanoes

Request volunteers to use Encarta to research the various types of volcanoes (strato- or composite volcanoes, cinder cones, shield volcanoes, and perhaps calderas). Direct them to download pictures of each type of volcano. Have students turn these pictures into flashcards. When they are done, ask these volunteers to deliver a short oral report on volcanoes to the rest of the class. At the end of the report, have them challenge the class to a game of "Name that Volcano." Every time they hold up a flashcard, classmates should identify it by type. (Tell the volunteers that they should be ready to provide the name and location of each volcano.)

Identifying Volcanic Rocks

The lesson on volcanoes provides a good opportunity to develop, or build upon, prior knowledge of the rocks that form Earth's crust. You might point out that rocks are composed of minerals: naturally occurring, inorganic, solid elements or compounds with a fixed atomic pattern. Name the three types of rocks--igneous, sedimentary, and metamorphic. Then focus student attention on igneous rocks, the type of rock associated with volcanoes. Direct students to use Encarta to prepare a photo display of igneous rocks, such as pumice, granite, basalt, and obsidian. If possible, have them bring in examples of these rocks for the rest of the class to

examine. You can expand upon this lesson, by having other groups of students prepare similar displays of the two other types of rocks. Use the displays as an opportunity to talk about the geology of your area.

Visiting Extraterrestrial Hot Spots

Encarta Encyclopedia 2000 includes a sidebar article on the 1998 discovery of the hottest spot in the solar system: Io's Pele Volcano. You might use this story as a chance to lead students into a virtual tour of extraterrestrial volcanoes. They can read the **Planetary Science** article and the articles about individual planets—especially Jupiter, the planet that holds Io in its gravitational force. With this background in mind, they can head into outer space via the Internet. Suggested sites appear at the end of this lesson. Use student tours of outer space to discuss what the presence of volcanoes tells scientists about other planets and moons in our solar system.

Reporting on Volcanic Eruptions

The Encyclopedia article **Volcanoes** has a number of sidebar articles on specific volcanic eruptions. Assign each of these sidebars to a group of students. Direct students to imagine they are on-the-spot TV reporters at each eruption. Based on the sidebars, have them develop spot news reports to the rest of the class.

For articles, maps, or other media pertaining to this topic, look up the following in Encarta® Encyclopedia:

Geology
Pacific Ocean
Plate Tectonics
Plume (volcano)
Volcano

Refer to the following items in Encarta® Interactive World Atlas for relevant information:

Volcanoes, Global Theme
Searching on Volcano will turn up many articles and media items

The following items in the Encarta® World English Dictionary® contain information about this topic:

Caldera
Cone
Crater
Dormant (volcano)
Extinct (volcano)
Fissure
Igneous
Lava
Lithosphere
Magma
Pumice
Pyroclastics
Rift
Subduction
Vulcanologist
Vulcanology

For additional online information and media, view the following Encarta articles:

Plate Tectonics
Volcano

For additional relevant information, visit the following Web sites:
Continental Drift [Royal Tyrrell Museum of Palaeontology]
http://tyrrell.magtech.ab.ca/tour/contdrft.html
The Royal Tyrrell Museum of Palaeontology presents information on the movement of the continents in the earth's history.
Volcano World
http://volcano.und.edu/
Volcano World presents basic information about volcanoes as well as news on recent eruptions; funded by NASA and hosted by the University of North Dakota.
Global Volcanism Program
http://www.volcano.si.edu/gvp/
The Global Volcanism Program at the Smithsonian Institution offers an electronic version of the monthly Bulletin of the Global Volcanism Network, with reports on current volcanic eruptions; a complete database of volcanoes; and other resources.